UNIT - III

Fixed Beams and Continuous Beams

1. Introduction to Statically Indeterminate Beams

- Beams with **more than three reaction components** cannot be solved by statics alone — they are **statically indeterminate beams** .
- Main types: **(i) propped cantilever, (ii) fixed (built-in) beam, (iii) continuous beam** .



Types of indeterminate beams

- Methods: **Castigliano II, consistent deformation, moment-area, three-moment theorem (Clapeyron), slope-deflection, moment distribution** .

2. Fixed Beams — Meaning and Features

- A **fixed beam (built-in or encastré beam)** has both ends rigidly fixed so that **slope and deflection at both ends are zero** .
- Reaction components: at each end vertical reaction, horizontal reaction, and **fixed end moment** → 6 unknowns; $D_s = 3$ (**2 for vertical loads** since $H = 0$).

Basis	Simply supported beam	Fixed beam
End conditions	free to rotate; $M = 0$ at ends	end rotation prevented; fixed end moments M_A, M_B
Max bending moment	$wL^2/8$ (UDL)	$wL^2/12$ (at supports) — **33% less**
Max deflection (UDL)	$5wL^4/384EI$	$wL^4/384EI$ — **1/5th**
Stiffness	less	more
Economy	larger sections	smaller sections
Settlement of support	no stresses	causes stresses (disadvantage)
Analysis	simple (equilibrium)	indeterminate

Advantages of fixed beams

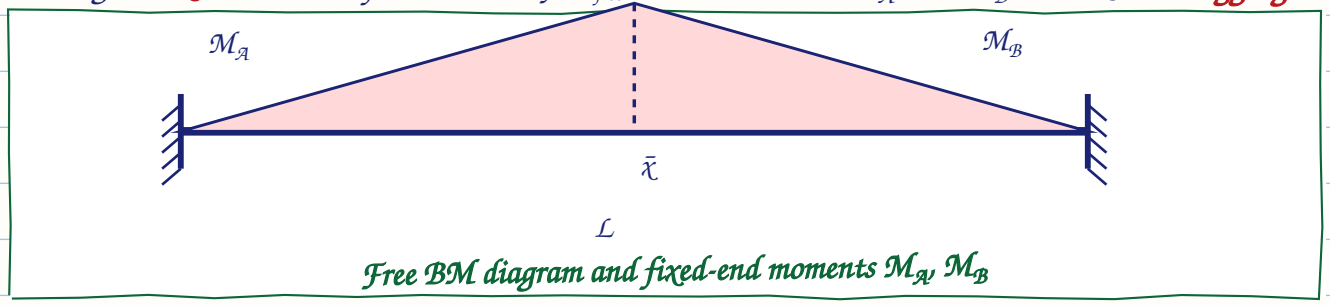
- Smaller maximum bending moment and deflection; more stiff and strong; economical section; more stable under load.

Disadvantages

- Fixed end moments need heavy end connections; **settlement, temperature** and rotation of supports produce extra stresses; analysis is difficult; fixed ends are hard to achieve in practice.

3. Fixed End Moments — Area-Moment Method

- Consider fixed beam AB, span L, constant EI. Let **a** = area of the free (simply supported) BM diagram, **$\bar{x}$** = distance of its centroid from A. End moments M_A and M_B are taken as **hogging**.



Free BM diagram and fixed-end moments M_A, M_B

- **Conditions** (slopes at A and B are the same, so change of slope = 0; deflection of B from tangent at A = 0):

1. Area of net BM diagram = 0: $a = (M_A + M_B)\,L/2$.
2. Moment of net M diagram about A = 0: $a\bar{x} = M_A L^2/6 + M_B L^2/3$ (moment of the trapezoid about A). Solving both conditions:

Fixed end moments (general)

$$M_A = 2a(2L - 3\bar{x})/L^2 \quad M_B = 2a(3\bar{x} - L)/L^2$$

- Check for central load: $a = PL^2/8$, $\bar{x} = L/2 \rightarrow M_A = M_B = PL/8$ ✓.
- **Reactions:** R_A = (free reaction at A) + $(M_A - M_B)/L$; R_B = (free reaction at B) − $(M_A - M_B)/L$.
- **Net BM at any section = free BM − fixed end moment diagram (trapezoid)**.

Procedure

Draw free BM diagram → Find area a and centroid $\bar{x}$ → Find M_A and M_B → Find reactions

Draw SFD and BMD → Locate max BM and contraflexure

→ If end moments are equal (symmetrical loading) $M_A = M_B = a/L$ (average height of the free BM diagram).

4. Fixed End Moments for Standard Cases

Loading on fixed beam of span L	M_A (fixed end A)	M_B (fixed end B)
UDL w over the whole span	$wL^2/12$	$wL^2/12$
Central point load P	$PL/8$	$PL/8$
Eccentric point load P at a from A, b from B	Pab^2/L^2	Pa^2b/L^2
Two equal loads P at $L/3$ and $2L/3$	$2PL/9$	$2PL/9$
Triangular load, zero at A to w at B (UVL)	$wL^2/30$	$wL^2/20$
Triangular load, w at A to zero at B	$wL^2/20$	$wL^2/30$
Triangular (symmetrical, peak w at centre)	$5wL^2/96$	$5wL^2/96$
UDL w over left half only (length $L/2$ from A)	$11wL^2/192$	$5wL^2/192$
Couple M_0 at distance a from A ($b = L - a$)	$M_0 b(2a - b)/L^2$	$M_0 a(2b - a)/L^2$

- For downward loads the end moments are **hogging**. For the couple the magnitudes are given (a clockwise couple is taken positive); one end is hogging and the other sagging.

Derivation 1 — UDL (a quick proof)

- Free BM $= wLx/2 - wx^2/2$, area $a = (2/3)\cdot L\cdot(wL^2/8) = wL^3/12$, $\bar{x} = L/2 \rightarrow M_A = 2a(2L - 1.5L)/L^2 =$ **$wL^2/12$**.

Derivation 2 — Eccentric point load

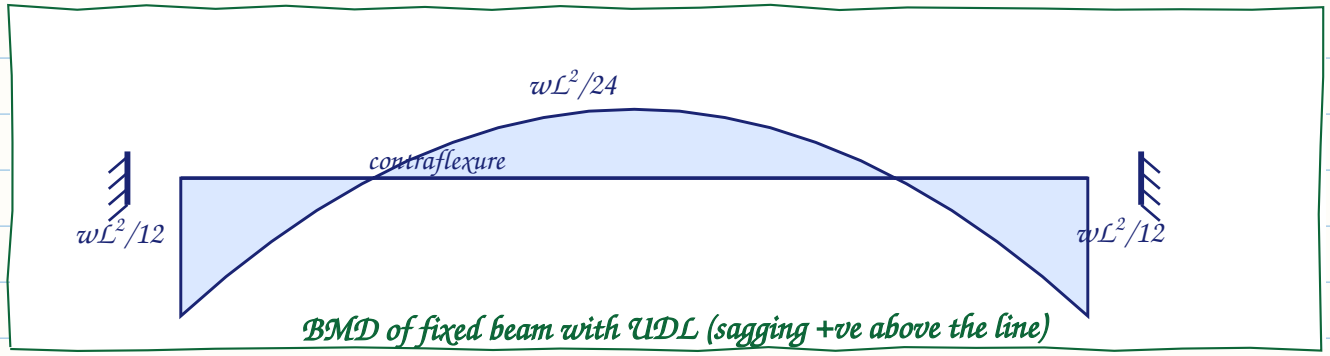
- Free BM is a triangle with peak Pab/L under the load: area $= \tfrac{1}{2} \times L \times (Pab/L) = Pab/2$; centroid from A: $\bar{x} = (L + a)/3$.
- $M_A = 2\,(Pab/2)(2L - (L + a))/L^2 =$ **Pab^2/L^2** and likewise $M_B =$ **Pa^2b/L^2** (the larger moment is at the end nearer to the load).

Derivation 3 — UVL (zero at A, w at B)

- $R_{A0} = wL/6$; $M(x) = wLx/6 - wx^3/6L$; area $a = wL^3/24$; $\bar{x} = 8L/15 \rightarrow M_A =$ **$wL^2/30$**, $M_B =$ **$wL^2/20$**.

5. Fixed Beam — Reactions, SFD and BMD (Worked Examples)

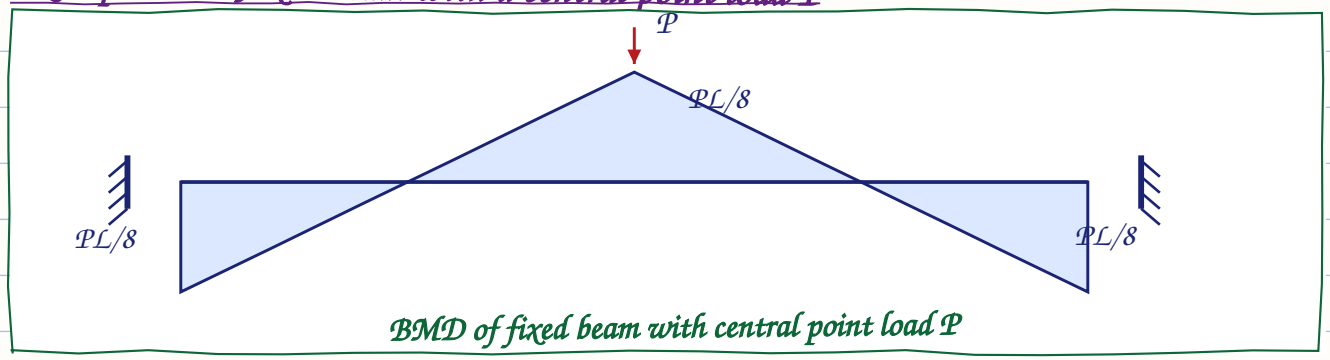
Example 1 — Fixed beam with UDL over the whole span



BMD of fixed beam with UDL (sagging +ve above the line)

- $M_A = M_B = wL^2/12$ (hogging); $R_A = R_B = wL/2$. Centre BM = $wL^2/8 - wL^2/12 =$ **$+ wL^2/24$** .
- **Points of contraflexure** at $x = L(1 \pm 1/\sqrt{3})/2 =$ **0.211 L** from each end.
- SFD: straight line from $+ wL/2$ at A to $- wL/2$ at B, zero at the centre.

Example 2 — Fixed beam with a central point load P



BMD of fixed beam with central point load P

- $M_A = M_B = PL/8$; $R_A = R_B = P/2$; net M at the centre = $PL/4 - PL/8 =$ **$+ PL/8$** . Contraflexure at $L/4$ from each end.
- SFD: $+ P/2$ constant in AC and $- P/2$ in CB (a jump of P at the load).

Example 3 — Numerical: fixed beam with UDL + central point load

Problem

A fixed beam of span 6 m carries a UDL of 20 kN/m over the entire span and a point load of 40 kN at the centre. Find the fixed end moments, reactions and the centre BM.

- UDL: $wL^2/12 = 20 \times 36/12 = 60$ kN·m; point load: $PL/8 = 40 \times 6/8 = 30$ kN·m.
- **$M_A = M_B = 60 + 30 = 90$ kN·m** (hogging).
- **$R_A = R_B = (20 \times 6 + 40)/2 = 80$ kN**.
- Free BM at centre = $wL^2/8 + PL/4 = 90 + 60 = 150$ kN·m → **net centre BM = 150 − 90 = 60 kN·m** (sagging).

Example 4 — Eccentric point load (numerical)

Problem

Fixed beam of span 6 m; P = 60 kN at 2 m from A. Find M_A, M_B, R_A, R_B and BM under the load.

- $a = 2$ m, $b = 4$ m: $M_A = P\,a\,b^2/L^2 = 60 \times 2 \times 16 / 36 =$ **53.33 kN·m** ; $M_B = P\,a^2\,b/L^2 = 60 \times 4 \times 4 / 36 =$ **26.67 kN·m** .

- $R_A = P\,b^2\,(3a + b)/L^3 = 60 \times 16 \times 10 / 216 =$ **44.44 kN**; $R_B = 60 - 44.44 =$ **15.56 kN**.
- BM under load $= R_A \times 2 - M_A = 88.89 - 53.33 =$ **+ 35.56 kN·m** (check from right: $15.56 \times 4 - 26.67 = 35.56$ √).

Example 5 — Number of point loads and combination

- Use **superposition**: sum the fixed end moments of each load separately, e.g., loads of 30 kN at 2 m and 20 kN at 4 m on a 6 m beam.
- Case 30 kN: $a = 2, b = 4 \rightarrow M_A = 30 \times 2 \times 16/36 = 26.67$; $M_B = 30 \times 4 \times 4/36 = 13.33$.
- Case 20 kN: $a = 4, b = 2 \rightarrow M_A = 20 \times 4 \times 4/36 = 8.89$; $M_B = 20 \times 16 \times 2/36 = 17.78$.
- **Total $M_A = 35.56$ kN·m, $M_B = 31.11$ kN·m**. free $R_{A0} = (30 \times 4 + 20 \times 2)/6 = 26.67$ kN; $R_A = R_{A0} + (M_A - M_B)/L = 26.67 + (35.56 - 31.11)/6 =$ **27.41 kN**; $R_B = 50 - 27.41 =$ **22.59 kN**.

Example 6 — Fixed beam with UVL (numerical)

- Span 6 m, load rising from 0 at A to 24 kN/m at B: $M_A = wL^2/30 = 24 \times 36/30 =$ **28.8 kN·m**; $M_B = wL^2/20 =$ **43.2 kN·m**.
- Total load $= ½ \times 24 \times 6 = 72$ kN; free $R_{A0} = wL/6 = 24$ kN; $R_A = 24 + (28.8 - 43.2)/6 =$ **21.6 kN**; $R_B = 72 - 21.6 =$ **50.4 kN**.

Example 7 — Fixed beam with a couple M_0

- Couple M_0 at the centre: $M_A = M_B = M_0/4$ (equal in magnitude — one end hogging, the other sagging); reactions $R_A = -R_B = 3M_0/2L$ (a couple of reactions).
- For a couple at a distance a from A: $M_A = M_0\,b\,(2a - b)/L^2$, $M_B = M_0\,a\,(2b - a)/L^2$ (clockwise couple positive).

6. Deflection of Fixed Beams

- Fixed beams are much stiffer. Deflection is obtained by the **net BM diagram** (moment-area) or **superposition** of the simple-beam deflection and the deflection due to end moments.

Loading	Max deflection (at centre)	Compare S.S.
UDL w over full span	$w\,L^4 / 384\,EI$	$5wL^4/384EI$ (5 times)
Central point load P	$P\,L^3 / 192\,EI$	$PL^3/48EI$ (4 times)
Eccentric load P (a from A, b from B)	under the load: $P\,a^3\,b^3 / 3\,E\,I\,L^3$	—

- **Derivation (UDL):** $\delta_{centre} = \delta$(simple) $- \delta$(due to end moments $M_0 = wL^2/12$) $= 5wL^4/384EI - (M_0\,L^2/8EI) = wL^4/EI\,(5/384 - 4/384) =$ **$wL^4/384EI$**.
- For an eccentric load the maximum deflection occurs in the **longer segment** (where the slope is zero); it is slightly more than the deflection under the load.

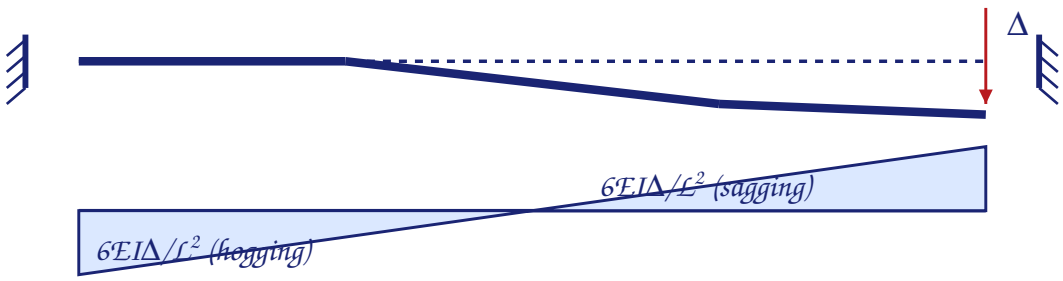
7. Effect of Sinking (Settlement) of Support

- If one support B of a fixed beam **sinks by Δ** (no rotation of ends), the beam deforms like a guided bar; **no load needed** — moments and shears arise from the displacement.
- Chord rotation $\psi = \Delta/L$.

Fixed end moments due to settlement Δ

$M_A = M_B = 6EI\Delta / L^2$ (A hogging, B sagging) (BM varies linearly)

Shear $= 12EI\Delta / L^3$ (constant) — R_A increases, R_B decreases by this amount



Fixed beam, support B sinks by Δ : beam shape and BMD

- **Derivation:** Slope-deflection: $M_{AB} = 2EI/L\,(2\theta_A + \theta_B - 3\psi)$ with $\theta_A = \theta_B = 0 \rightarrow M_{AB} = -6EI\Delta/L^2$; similarly $M_{BA} = -6EI\Delta/L^2$.
- The final moments = moments due to loads + moments due to settlement (superposition).

Example 8 — Numerical (settlement)

Problem

Fixed beam, span 6 m, $EI = 30000$ kN·m², right support sinks by 10 mm. Find end moments and reaction change.

- $M = 6EI\Delta/L^2 = 6 \times 30000 \times 0.01 / 36 =$ **50 kN·m** .
- $\Delta R = 12EI\Delta/L^3 = 12 \times 30000 \times 0.01 / 216 =$ **16.67 kN**.
- If a UDL $w = 20$ kN/m also acts: $M_A = wL^2/12 + 50 = 60 + 50 = 110$ kN·m (hogging at A); $M_B = 60 - 50 = 10$ kN·m (hogging at B).

Observation

- Settlement **increases** the moment at one end and **reduces** it at the other — a serious point in design of indeterminate beams.

8. Effect of Rotation of a Support

- If the end A rotates by θ (clockwise) with B still fixed and no relative deflection:

Moments due to rotation θ of support A

$M_A = 4EI\theta / L$, $M_B = 2EI\theta / L$ (carry-over factor ½)

Shear = $6EI\theta / L^2$ (R_A and R_B equal and opposite)

- If **both ends rotate** by θ_A and θ_B: $M_A = (2EI/L)(2\theta_A + \theta_B)$, $M_B = (2EI/L)(2\theta_B + \theta_A)$.
- **Example:** $L = 5$ m, $EI = 20000$ kN·m², end A rotates 0.002 rad: $M_A = 4 \times 20000 \times 0.002 / 5 =$ **32 kN·m** ; $M_B =$ **16 kN·m** ; shear $= 6 \times 20000 \times 0.002/25 =$ **9.6 kN**.
- A simply supported (pin) end has **zero** moment after rotation; so rotation of a support matters only for a **fixed** end.

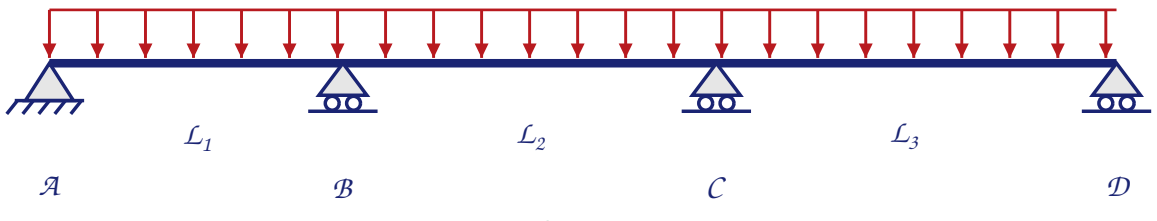
9. Propped Cantilever — Standard Results

Loading (A fixed, B propped)	R_B	M_A
UDL w over full span	$3wL/8$	$wL^2/8$ (hogging)
Central point load P	$5P/16$	$3PL/16$
Point load P at a from A	$P\,a^2\,(3L - a) / 2L^3$	$P\,a\,b\,(L + b) / 2L^2$

- Prop reaction is found from **zero deflection at the prop** (Unit II methods).

10. Continuous Beams — Introduction

- A **continuous beam** rests on **more than two supports** ; it is continuous over the intermediate supports.
- Degree of indeterminacy = (number of reaction components) − 3 = **n − 2** for n simple supports.



Continuous beam with three spans

Advantages

- Smaller BM than a series of S.S. beams; economical; fewer expansion joints; more rigid; smooth riding (bridges); redistribution of moments on overload.

Disadvantages

- Analysis is difficult; **settlement of supports** changes the moments; **temperature** ; and a hogging moment at the supports needs reinforcement on the top.

Features of BMD

- **Hogging** BM over the interior supports and **sagging** BM in the middle of the spans; points of contraflexure exist in every span.

11. Clapeyron's Theorem of Three Moments

- Gives a relation between the **hogging moments at three consecutive supports**. Each span has its own free BM diagram.

Theorem of three moments (supports at same level, EI constant, hogging +ve)

$$M_A L_1 + 2 M_B (L_1 + L_2) + M_C L_2 = 6 a_1 \bar{x}_1 / L_1 + 6 a_2 \bar{x}_2 / L_2$$

- a_1, a_2 = areas of free BM diagrams of spans AB and BC; $\bar{x}_1$ = distance of centroid of a_1 from **A** (the left support); $\bar{x}_2$ = distance of centroid of a_2 from **C** (the right support).
- **Standard right-hand sides ($6 a \bar{x} / L$):**

Load on a span of length L	$6 a \bar{x} / L$
UDL w	$w L^3 / 4$
Central point load P	$3 P L^2 / 8$
Point load P at a from the end considered ($b = L - a$)	$P a b (L + a)/L$ (from the left end); $P a b (L + b)/L$ (from the right end)
UVL (zero at one end, w at the other)	$2 w L^3/15$ ($\bar{x}$ measured from the zero end); $7 w L^3/60$ ($\bar{x}$ measured from the max end)

With support settlement

- If B is lower than A and C by δ_1 and δ_2: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = 6a_{1\bar{x}1}/L_1 + 6a_{2\bar{x}2}/L_2 - 6EI\,(\delta_1/L_1 + \delta_2/L_2)$.
- **Fixed end A:** add an imaginary span of zero length → $2 M_A L_1 + M_B L_1 = 6 a_1 \bar{x}_1 / L_1$.
- **Overhang:** the moment at the support = (load × distance) known; use it as M_A in the equation.

Procedure

Draw free BM diagrams → Write 3-moment equation for every pair of adjacent spans → Solve for support moments → Reactions

SFD and BMD

1. Number of three-moment equations = number of unknown support moments (= D_s).
2. Solve simultaneous equations; find reactions of every span: R = (free reaction) ± $(M_B - M_A)/L$.
3. Draw SFD, then BMD (free BM + support-moment trapezoid).

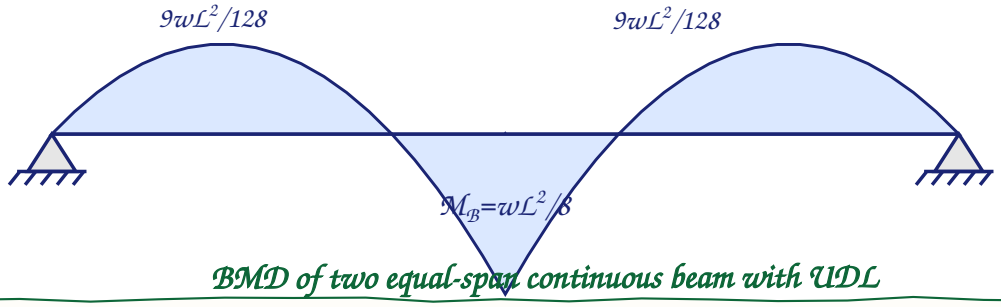
12. Continuous Beam — Solved Examples

Example 9 — Two spans with UDL

Problem

Continuous beam ABC, simply supported at A and C: AB = 4 m, BC = 6 m, UDL 20 kN/m on the whole length. Find the support moment and reactions.

- $M_A = M_C = 0$. Equation: $2\,M_B\,(4 + 6) = w\,(L_1^3 + L_2^3)/4 = 20\,(64 + 216)/4 = 1400 \rightarrow$ **M_B = 70 kN·m (hogging)**.
- $R_A = wL_1/2 - M_B/L_1 = 40 - 17.5 =$ **22.5 kN**; $R_C = 60 - 70/6 =$ **48.33 kN**; **R_B = 200 − 22.5 − 48.33 = 129.17 kN**.
- Max sagging in AB: $x = 22.5/20 = 1.125$ m from A → $M = 22.5 \times 1.125 - 10 \times 1.125^2 =$ **12.66 kN·m**.
- Max sagging in BC: $x = 48.33/20 = 2.417$ m from C → $M = 48.33 \times 2.417 - 10 \times 2.417^2 =$ **58.4 kN·m**.



BMD of two equal-span continuous beam with UDL

- The figure shows the equal-span case: $M_B = wL^2/8$ and sagging $9wL^2/128$.

Example 10 — Two equal spans with central point loads

- $L = 6$ m each, $P = 60$ kN at the centre of each span: $4\,M_B\,L = 2 \times 3PL^2/8 \rightarrow$ **$M_B = 3PL/16 = 3 \times 60 \times 6/16 = 67.5$ kN·m**.
- $R_A = P/2 - M_B/L = 30 - 11.25 =$ **18.75 kN** $= R_C$; $R_B = 120 - 37.5 =$ **82.5 kN**.
- BM under each load = $18.75 \times 3 =$ **56.25 kN·m** (sagging).

Example 11 — Beam fixed at A (two spans)

Problem

Beam ABC: A fixed, B and C rollers; AB = BC = 6 m; UDL 20 kN/m on both spans. Find the support moments and reactions.

- Fixed end A (imaginary zero span): $2\,M_A\,(6) + M_B\,(6) = w\,L^3/4 = 1080 \rightarrow$ **$2\,M_A + M_B = 180$**.
- Equation at B ($M_C = 0$): $6\,M_A + 24\,M_B = 1080 + 1080 \rightarrow$ **$M_A + 4\,M_B = 360$**.
- Solving: **M_A = 51.43 kN·m, M_B = 77.14 kN·m** (both hogging).
- Span AB: $R_A = 60 + (M_A - M_B)/6 =$ **55.71 kN**; R_B(left) = 64.29 kN. Span BC: $R_C = 60 -$

$77.14/6 = \mathbf{47.14\ kN}$; R_B(right) = 72.86 kN.

- $\mathbf{R_B = 137.14\ kN}$; check: 55.71 + 137.14 + 47.14 = 240 = total load √.

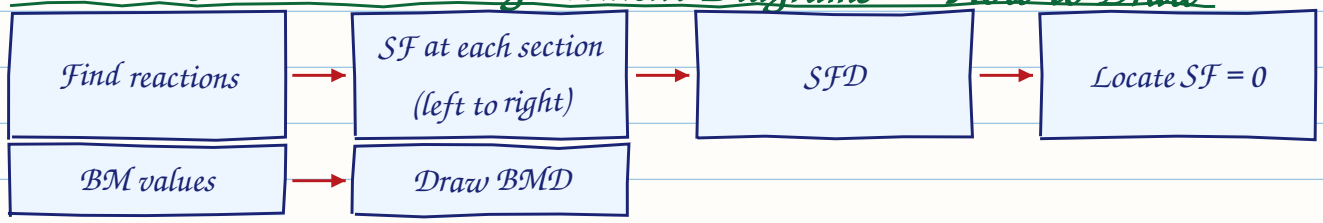
Example 12 — Effect of settlement

Problem

Two equal spans L = 6 m, UDL w = 20 kN/m, EI = 30000 kN·m²; the middle support B settles by 10 mm. Find M_B.

- $4\,M_B\,L = w\,L^3/2 - 6EI\Delta\,(2/L) \rightarrow M_B = wL^2/8 - 3EI\Delta/L^2 = 90 - 25 = \mathbf{65\ kN{\cdot}m}$.
- Without settlement M_B = 90 kN·m; settlement **reduces** the hogging moment at B (and increases the sagging in the spans).

13. Shear Force and Bending Moment Diagrams — How to Draw



- **SFD:** starts with R_A, drops by load, jumps up at a reaction; slope = − w for UDL.
- **BMD:** area under SFD gives change in BM; **BM is max where SF = 0**.
- **Points of contraflexure:** where BM changes sign (BM = 0).
- For fixed beams: BMD = free BM diagram (parabola / triangle) **minus** the end-moment trapezoid; for continuous beams = free BM + support-moment line.

Sign conventions

- Sagging BM +ve (draw above/below as per your textbook — mention clearly); hogging −ve.
- SF +ve when the left part tends to move up relative to the right.

14. Comparison — Fixed vs Continuous vs Propped Cantilever

Basis	Propped cantilever	Fixed beam	Continuous beam
Supports	fixed + roller	fixed + fixed	three or more supports
D_s	1	3 (2)	$n - 2$
UDL max BM	$wL^2/8$	$wL^2/12$	$wL^2/8$ at support (equal spans)
Best method	Castigliano II	area-moment	three-moment, slope deflection

14A. Three-Span Continuous Beam — Equal Spans with UDL

- Supports A, B, C, D; spans L each; symmetry gives $M_B = M_C = M$; $M_A = M_D = 0$.
- Equation at B: $2M(2L) + ML = 2 \times wL^3/4 \rightarrow 5ML = wL^3/2 \rightarrow$ **$M_B = M_C = wL^2/10$** .
- Numerical: $L = 5$ m, $w = 20$ kN/m → **$M = 20 \times 25/10 = 50$ kN·m (hogging)** .
- $R_A = wL/2 - M/L = 50 - 10 =$ **40 kN** $= R_D$; $R_B = R_C = (300 - 80)/2 =$ **110 kN** $(= 1.1\,wL)$.
- Sagging BM: end span $x = 40/20 = 2$ m → $40 \times 2 - 10 \times 4 =$ **40 kN·m** ; middle span centre: $wL^2/8 - M = 62.5 - 50 =$ **12.5 kN·m** .

Equal spans, UDL	Two-span	Three-span	S.S. span
Support moment	$wL^2/8$	$wL^2/10$	0
End-support reaction	$3wL/8$	$2wL/5$	$wL/2$
Interior reaction	$5wL/4$	$11wL/10$	—

14B. Continuous Beam with Overhang and Mixed Loads

Problem

Beam on supports A, B, C with spans AB = BC = 6 m and an overhang CD = 2 m carrying a point load of 30 kN at D. A UDL of 10 kN/m acts on AB and BC. Find the support moments and reactions.

- **Overhang moment at C** (known from statics): $M_C = -30 \times 2 = -60$ kN·m, i.e. **hogging 60 kN·m** .
- Equation at B: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = wL_1^3/4 + wL_2^3/4 \rightarrow 0 + 24M_B + 360 = 540 + 540 \rightarrow$ **$M_B = (1080 - 360)/24 = 30$ kN·m** (hogging).
- Span AB: $R_A = 30 + (0 - 30)/6 =$ **25 kN**; R_B(left) $= 30 + 5 = 35$ kN.
- Span BC (hogging 30 at B and 60 at C): R_B(right) $= 30 + (30 - 60)/6 = 25$ kN; R_C(span) $= 30 + 5 = 35$ kN.
- **$R_B = 35 + 25 = 60$ kN**; **$R_C = 35 + 30$ (overhang load) $= 65$ kN**; check: $25 + 60 + 65 = 150 = 120 + 30$ √.

→ Treat the overhang as a cantilever: its moment at the support is fixed by statics and used in the equation.

14C. Propped Cantilever — More Cases (A fixed, B propped)

- **Couple M_0 applied at the prop B:** $R_B = 3M_0/2L$, $M_A = M_0/2$ (carry-over).
- **UDL over the whole span:** $R_B = 3wL/8$, $M_A = wL^2/8$ (hogging); BM at $3L/8 = 9wL^2/128$.
- **Point load at a from fixed end A:** $R_B = P\,a^2\,(3L - a)/2L^3$.
- **Check with $a = L/2$:** $R_B = 5P/16$ √.
- Example: $L = 6$ m, $P = 48$ kN at 2 m from A: $R_B = 48 \times 4 \times (18 - 2)/(2 \times 216) =$ **7.11 kN**; R_A

$= 48 - 7.11 = 40.89$ kN; $M_A = P\,a - R_B\,L = 96 - 42.67 =$ **53.33 kN·m** (hogging). Check: $P\,a\,b\,(L + b)/2L^2 = 48 \times 2 \times 4 \times 10/72 = 53.33$ √.

14D. Redistribution of Moments in Continuous Beams

- In reinforced concrete and steel design, continuous beams develop **plastic hinges** at supports on overload; the BM diagram then **redistributes** — moments at supports reduce and span moments increase.
- This is why indeterminate structures have **reserve strength**; codes allow limited (up to 15–30%) redistribution.
- Elastic analysis (three moments, slope-deflection, moment distribution) is still the starting point.

14E. Common Mistakes in Fixed and Continuous Beam Problems

- Confusing **a** (area of free BM) and $\bar{x}$ (distance from A) — recheck with a central load where $M = PL/8$.
- Using wrong sign for settlement — settlement of the middle support **reduces** hogging at that support.
- For the three-moment theorem: **$\bar{x}$ is measured from the outer end** (left end of span 1, right end of span 2).
- Forgetting the fixed-end condition (imaginary zero-length span).
- Not adding the **overhang moment** at the support.
- Drawing the BMD without marking the **maximum moments and points of contraflexure**.
- Not checking Σ reactions = total load.

14F. Practice Questions with Answers

No.	Problem	Answer
1	Fixed beam 8 m, UDL 15 kN/m	$M_A = M_B = 80$ kN·m; centre +40 kN·m
2	Fixed beam 5 m, central load 40 kN	$M_A = M_B = 25$ kN·m; centre +25 kN·m
3	Fixed beam 6 m, load 36 kN at 2 m from A	$M_A = 32$ kN·m, $M_B = 16$ kN·m
4	Continuous beam, 2 equal spans of 4 m, UDL 25 kN/m	$M_B = 50$ kN·m; $R_B = 125$ kN
5	Fixed beam 4 m, EI = 20000, end sinks 5 mm	$M = 6EI\Delta/L^2 = 37.5$ kN·m
6	Fixed beam 6 m, UVL 0 to 30 kN/m (A to B)	$M_A = 36$ kN·m; $M_B = 54$ kN·m

15. Quick Revision — Unit III

Formulae

Fixed beam: $M_A = 2a(2L - 3\bar{x})/L^2$, $M_B = 2a(3\bar{x} - L)/L^2$ | UDL: $wL^2/12$ | Central load: $PL/8$ | Eccentric: Pab^2/L^2, Pa^2b/L^2 | Settlement: $6EI\Delta/L^2$ | Rotation θ: $4EI\theta/L$, $2EI\theta/L$ | Three moments: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = 6a_{1}\bar{x}_1/L_1 + 6a_{2}\bar{x}_2/L_2$ | Equal spans UDL: $M_B = wL^2/8$, $R_B = 5wL/4$.

Possible exam questions

1. Derive the fixed end moments for a fixed beam of span L with an eccentric point load.
2. Draw SFD and BMD for a fixed beam with UDL/central load; locate points of contraflexure.
3. Explain the effect of sinking of a support on the fixed end moments (derive).
4. State and derive Clapeyron's theorem of three moments.
5. Analyse a two-span continuous beam with UDL and point loads; draw SFD and BMD.
6. A continuous beam with a settlement of support (numerical).